



Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

FORT STREET HIGH SCHOOL

**2013**

**HIGHER SCHOOL CERTIFICATE COURSE**

**ASSESSMENT TASK 2**

# Mathematics Extension 1

TIME ALLOWED: 1½ HOURS  
(PLUS 5 MINUTES READING TIME)

| Outcomes Assessed                                                                               | Questions | Marks |
|-------------------------------------------------------------------------------------------------|-----------|-------|
| Determines integrals by reduction to a standard form through a given substitution.              | 1, 2, 3   |       |
| Manipulates algebraic expressions to solve problems involving inverse functions                 | 4, 5, 6   |       |
| Synthesises mathematical solutions to harder problems and communicates them in appropriate form | 7, 8      |       |

| Question | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | Total | % |
|----------|----|----|----|----|----|----|----|----|-------|---|
| Marks    | /8 | /8 | /9 | /8 | /9 | /8 | /8 | /7 | /65   |   |

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

Question 1 8 marks

- a. Sketch, on the same diagram, the curves  $y = \sin x$  and  $y = \sin^2 x$  for  $0 \leq x \leq \pi$  2 marks
- b. Find the area bounded by the curves in the domain. 2 marks
- c. Show that  $\sin^2 x - \sin^4 x = \frac{1}{4} \sin^2 2x$  2 marks
- d. Find the volume generated when the area from part b is rotated about the  $x$  axis. 2 marks

Question 2 8 marks

- a.  $\int x\sqrt{3-x} \, dx$  using the substitution  $u = 3-x$  3 marks
- b. Without the use of a calculator, give the exact value of:  $\tan\left\{\sin^{-1}\left(\frac{12}{13}\right)\right\}$  2 marks  
(Show working)
- c. Find the inverse function of  $f(x) = (x-1)^3$  and sketch both curves. 3 marks

Question 3 9 marks

- a. Evaluate  $\int_0^{\frac{\pi}{2}} \cos^2 4x \, dx$  4 marks
- b. Show that  $\int_0^{\frac{1}{4}} \frac{3x}{(1+4x)^3} \, dx = \frac{3}{128}$ , using the substitution  $u = 1+4x$  5 marks

Question 4 8 marks

a.

Differentiate the following with respect to  $x$

i.  $y = \sin^{-1}(x^2)$  2 marks

ii.  $y = \ln(\tan^{-1} x)$  2 marks

b.

Show that if  $y = e^{\cos \operatorname{csc} x}$ , then  $\frac{dy}{dx} = -\cot x \cos \operatorname{csc} x e^{\cos \operatorname{csc} x}$  2 marks

c.

Find the equation of the normal to the curve  $y = \tan^{-1} x$  at the point where  $x = 1$ .

2 marks

Question 5 9 marks

a.

Find the primitive functions of

i.  $\frac{x}{4+x^2}$  2 marks

ii.  $\frac{-1}{\sqrt{1-4x^2}}$  2 marks

b.

i. Show that  $\frac{2x^2+5}{(4+x^2)(1+x^2)} = \frac{1}{4+x^2} + \frac{1}{1+x^2}$  1 mark

ii. Hence evaluate  $\int_0^1 \frac{2x^2+5}{(4+x^2)(1+x^2)} dx$  correct to 2 decimal places 2 marks

c.

The area bounded by the curve  $y = \frac{1}{\sqrt{4+x^2}}$ , the  $x$  axis and the lines  $x = -2$  2 marks

and  $x = 2$  is rotated about the  $x$  axis. Find the volume of the solid generated.

Question 6 8 marks

a.

i. Find the domain and range of  $y = 5 \sin^{-1} \frac{x}{3}$

2 marks

ii. Sketch the curve

2 marks

b.

Calculate leaving your answer in exact form

4 marks

$$\int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{\sqrt{1-16x^2}}$$

Question 7 8 marks

a.

i. If  $y = \tan^{-1} x$ , express  $x$  as a function of  $y$

1 mark

ii. Prove that

4 marks

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

b.

Find the function whose differential is

3 marks

$$\frac{x^2+12}{x^2+9}$$

Question 8 7marks

a.

On the same set of axes sketch

2 marks

$$f(x) = \sin^{-1} x \text{ and } g(x) = \cos^{-1} x.$$

b.

If  $h(x) = \sin^{-1} x \cos^{-1} x$ , find  $h'(x)$ .

1 mark

c.

Sketch  $h(x)$ . Find and label the stationary point.

4 marks

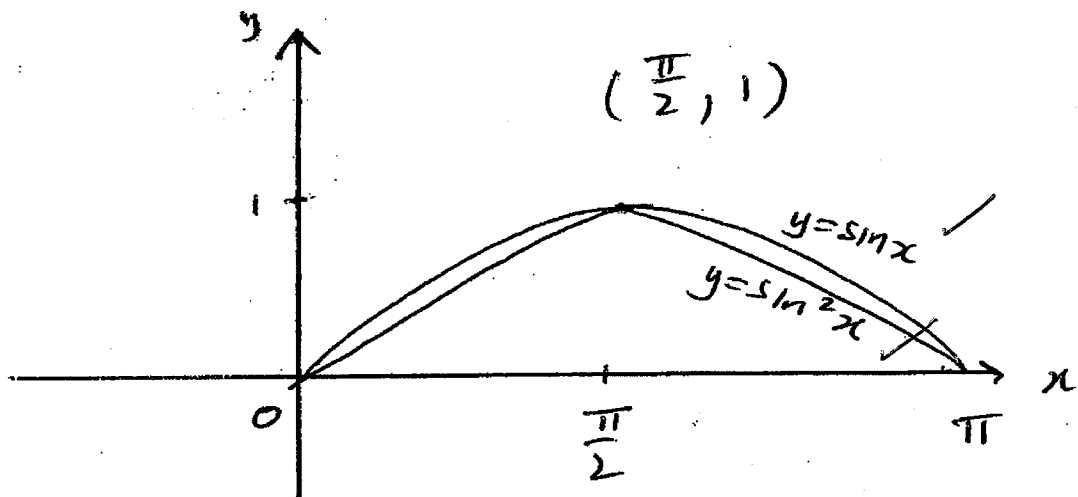
# Assessment task 2

## Extension 1

2013

### Question 1

a.



b.

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi} \sin x - \sin^2 x \, dx \\
 &= -[\cos x]_0^{\pi} - \frac{1}{2} \int_0^{\pi} 1 - \cos 2x \, dx \\
 &= -[\cos x]_0^{\pi} - \frac{1}{2} \left[ x - \frac{1}{2} \sin 2x \right]_0^{\pi} \\
 &= -(-1-1) - \frac{1}{2} [(\pi-0) - (0-0)] \\
 &= 2 - \frac{\pi}{2} \text{ units}^2
 \end{aligned}$$

c.

$$\begin{aligned}
 \text{LHS} &= \sin^2 x - \sin^4 x \\
 &= \sin^2 x (1 - \sin^2 x) \\
 &= \sin^2 x \times \cos^2 x \\
 &= (\sin x \cos x)^2 \\
 &= \left( \frac{1}{2} \sin 2x \right)^2 \\
 &= \frac{1}{4} \sin^2 2x
 \end{aligned}$$

d.

$$\begin{aligned}
 V &= \pi \int_0^{\pi} \sin^2 x - \sin^4 x \, dx \\
 &= \pi \int_0^{\pi} \frac{1}{4} \sin^2 2x \, dx \\
 &= \frac{\pi}{4} \left[ \frac{1}{2} (1 - \cos 4x) \right]_0^{\pi} \\
 &= \frac{\pi}{8} \left[ x - \frac{1}{4} \sin 4x \right]_0^{\pi} \\
 &= \frac{\pi}{8} [\pi - 0 - (0 - 0)] \\
 &= \frac{\pi^2}{8} \text{ units}^3
 \end{aligned}$$

Question 2

a.

$$I = \int x\sqrt{3-x} \, dx$$

$$u = 3 - x$$

$$du = -dx$$

$$dx = -du$$

$$I = \int (3-u) \times \sqrt{u} \times -du$$

$$= -\int (3-u)u^{\frac{1}{2}} \, du$$

$$= \int (u-3)u^{\frac{1}{2}} \, du$$

$$= \int u^{\frac{3}{2}} - 3u^{\frac{1}{2}} \, du$$

$$= \frac{2u^{\frac{5}{2}}}{5} - 2u^{\frac{3}{2}} + C$$

$$= \frac{2}{5}(3-x)^{\frac{5}{2}} - 2(3-x)^{\frac{3}{2}} + C$$

b.

Let

$$\alpha = \sin^{-1}\left(\frac{12}{13}\right)$$

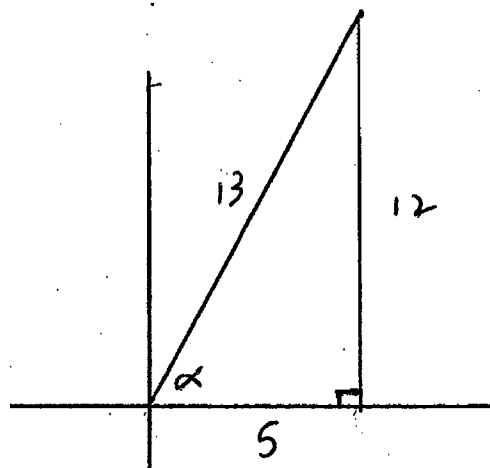
$$\sin \alpha = \frac{12}{13}$$

from the diagram

$$\tan \alpha = \frac{12}{5}$$

$$\text{since } \alpha = \sin^{-1}\left(\frac{12}{13}\right)$$

$$\text{then } \tan\left\{\sin^{-1}\left(\frac{12}{13}\right)\right\} = \frac{12}{5}$$

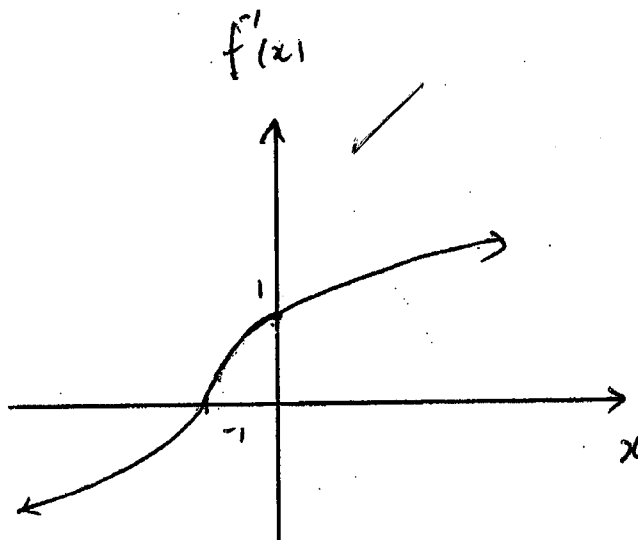
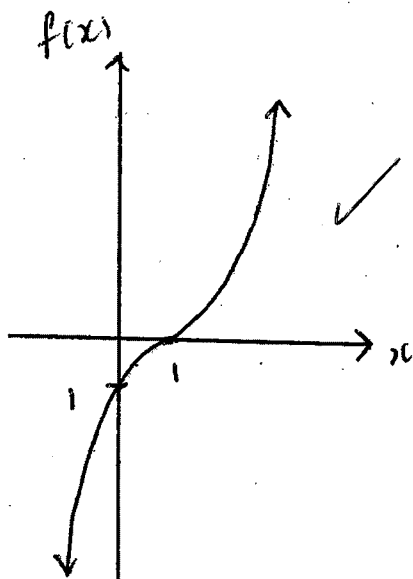


c.

$$f(x): y = (x-1)^3$$

$$f^{-1}(x): x = (y-1)^3$$

$$y = x^{\frac{1}{3}} + 1$$



Comments :

Question 3

a.

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \cos^2 4x \, dx \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 8x + 1 \, dx \quad \checkmark \\
 &= \frac{1}{2} \left[ \frac{1}{8} \sin 8x + x \right]_0^{\frac{\pi}{2}} \quad \checkmark \\
 &= \frac{1}{2} \left[ \frac{1}{8} \sin 4\pi + \frac{\pi}{2} - (0 - 0) \right] \quad \checkmark \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 \cos 8x &= \cos^2 4x - \sin^2 4x \\
 &= \cos^2 4x - (1 - \cos^2 4x) \\
 &= 2\cos^2 4x - 1 \\
 \frac{1}{2}(\cos 8x + 1) &= \cos^2 4x \quad \checkmark
 \end{aligned}$$

b.

Let

$$\begin{aligned}
 I &= \int_0^1 \frac{3x}{(1+4x)^3} \, dx \\
 &= \frac{3}{16} \int_1^2 \frac{3(u-1)}{4} \times \frac{1}{u^3} \times \frac{du}{4} \quad \checkmark \\
 &= \frac{3}{16} \int_1^2 \frac{u-1}{u^3} \, du \\
 &= \frac{3}{16} \int_1^2 u^{-2} - u^{-3} \, du \\
 &= \frac{3}{16} \left[ -u^{-1} + \frac{u^{-2}}{2} \right]_1^2 \quad \checkmark \\
 &= \frac{3}{16} \left[ \frac{1}{2u^2} - \frac{1}{u} \right]_1^2 \\
 &= \frac{3}{16} \left[ -\frac{1}{2} + \frac{1}{8} - \left( -1 + \frac{1}{2} \right) \right] \\
 &= \frac{3}{128} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 u &= 1 + 4x & u - 1 &= 4x & x &= \frac{1}{4} & u &= 2 \\
 \frac{du}{dx} &= 4 & 3\left(\frac{u-1}{4}\right) &= 3x & x &= 0 & u &= 1 \\
 dx &= \frac{du}{4} \quad \checkmark
 \end{aligned}$$

Question 4

a.

i.

$$y = \sin^{-1}(x^2)$$

$$y' = \frac{1}{\sqrt{1-(x^2)^2}} \times 2x$$

$$y' = \frac{2x}{\sqrt{1-x^4}}$$

ii.

$$y = \ln(\tan^{-1} x)$$

$$y' = \frac{1}{1+x^2} \div \tan^{-1} x$$

$$y' = \frac{1}{(1+x^2)\tan^{-1} x}$$

b.

$$y = e^{\operatorname{cosec} x}$$

$$y = e^u \quad \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = e^u \times \operatorname{cosec} x \cot x$$

$$\frac{dy}{dx} = \operatorname{cosec} x \cot x e^{\operatorname{cosec} x}$$

Let

$$u = \operatorname{cosec} x$$

$$u = \frac{1}{\sin x}$$

$$\frac{du}{dx} = \frac{\cos x}{\sin^2 x}$$

$$\frac{du}{dx} = \operatorname{cosec} x \cot x$$

c.

When  
 $x = 1$

$$y = \tan^{-1}(1) = \frac{\pi}{4}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

$$f'(1) = \frac{1}{2}$$

the normal has gradient

$$m = -2$$

$$y - \frac{\pi}{4} = -2(x-1)$$

$$y - \frac{\pi}{4} = -2x + 2$$

$$y = -2x + 2 - \frac{\pi}{4}$$

Question 5

a.

i.

$$\begin{aligned} & \int \frac{x}{4+x^2} dx \\ &= \frac{1}{2} \int \frac{2x}{4+x^2} dx \\ &= \frac{1}{2} \ln(4+x^2) + C \end{aligned}$$

ii.

$$\begin{aligned} & \int \frac{-1}{\sqrt{1-4x^2}} dx \\ &= \frac{1}{2} \int \frac{-1}{\sqrt{\frac{1}{4}-x^2}} dx \\ &= \frac{1}{2} \int \frac{-1}{\sqrt{\left(\frac{1}{2}\right)^2-x^2}} dx \\ &= \frac{1}{2} \cos^{-1}(2x) + C \end{aligned}$$

b.

i.

$$\begin{aligned} RHS &= \frac{1}{4+x^2} + \frac{1}{1+x^2} \\ &= \frac{1+x^2+4+x^2}{(4+x^2)(1+x^2)} \\ &= \frac{2x^2+5}{(4+x^2)(1+x^2)} \\ &= LHS \end{aligned}$$

ii.

$$\begin{aligned} & \int_0^1 \frac{2x^2+5}{(4+x^2)(1+x^2)} dx \\ &= \int_0^1 \frac{1}{4+x^2} + \frac{1}{1+x^2} dx \\ &= \left[ \tan^{-1} \frac{x}{2} + \tan^{-1} x \right]_0^1 \\ &= \tan^{-1} \frac{1}{2} + \tan^{-1} 1 - (\tan^{-1} 0 + \tan^{-1} 0) \\ &= 1.25 \end{aligned}$$

c.

$$\begin{aligned} V &= \pi \int_{-1}^2 \frac{1}{4+x^2} dx \\ &= 2\pi \left[ \tan^{-1} \left( \frac{x}{2} \right) \right]_0^2 \\ &= 2\pi [\tan^{-1} 1 - \tan^{-1} 0] \\ &= 2\pi \times \frac{\pi}{4} \\ &= \frac{\pi^2}{2} \text{ units}^3 \end{aligned}$$

Question 6

a.

i. Domain

$$-1 \leq \frac{x}{3} \leq 1$$

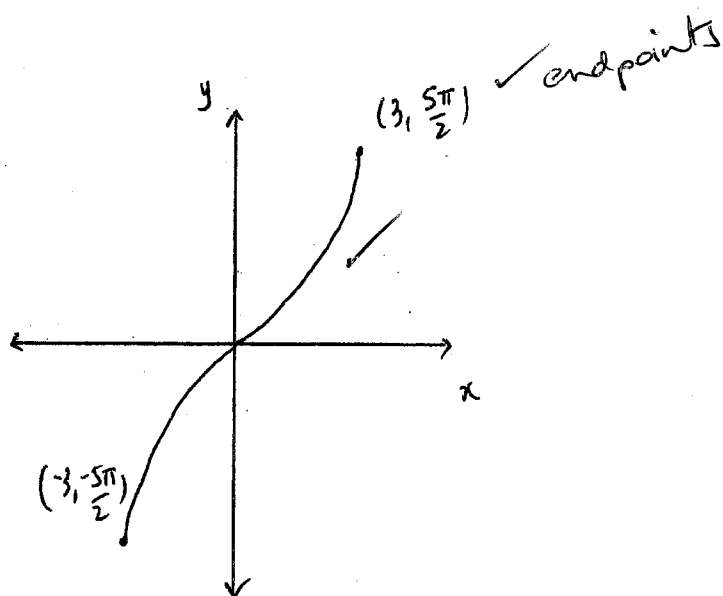
$$-3 \leq x \leq 3$$

Range

$$-\frac{\pi}{2} \leq \frac{y}{5} \leq \frac{\pi}{2}$$

$$-\frac{5\pi}{2} \leq y \leq \frac{5\pi}{2}$$

ii.



b.

$$\int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{\sqrt{1-16x^2}}$$

$$= \frac{1}{4} \int_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}} \frac{dx}{\sqrt{\left(\frac{1}{4}\right)^2 - x^2}}$$

$$= \frac{1}{4} \left[ \sin^{-1}(4x) \right]_{\frac{1}{8}}^{\frac{\sqrt{3}}{8}}$$

$$= \frac{1}{4} \left[ \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2} \right]$$

$$= \frac{1}{4} \left( \frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$= \frac{\pi}{24}$$

# Comments

## Question 7

a.

$$y = \tan^{-1} x$$

$$x = \tan y$$

b.

$$y = \tan^{-1} x$$

$$x = \tan y$$

$$\frac{dx}{dy} = \sec^2 y$$

$$\frac{dx}{dy} = 1 + \tan^2 y$$

$$\frac{dx}{dy} = 1 + x^2$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

c.

$$\int \frac{x^2+12}{x^2+9} dx$$

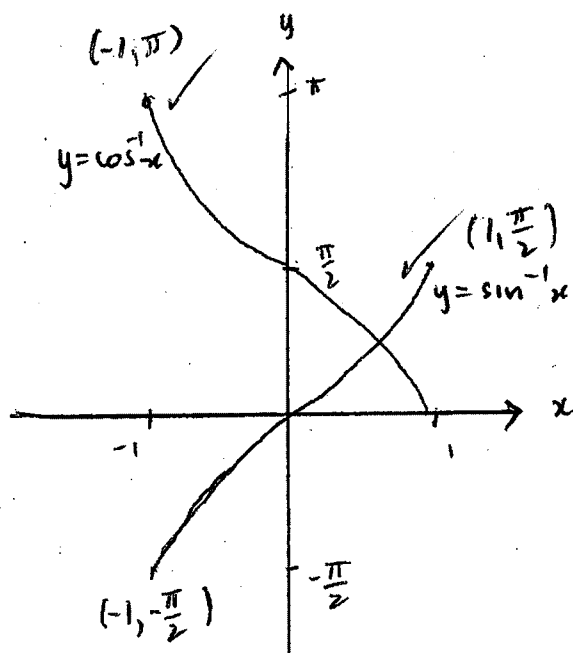
$$= \int \frac{x^2+9+3}{x^2+9} dx$$

$$= \int 1 + \frac{3}{x^2+9} dx$$

$$= x + 3 \tan^{-1} \frac{x}{3} + C$$

Question 8

a.



b.

$$h(x) = \sin^{-1} x \cos^{-1} x$$

$$u(x) = \sin^{-1} x \quad v(x) = \cos^{-1} x$$

$$u'(x) = \frac{1}{\sqrt{1-x^2}} \quad v'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$h'(x) = \sin^{-1} x \times \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x \times \frac{1}{\sqrt{1-x^2}}$$

$$h'(x) = \frac{\cos^{-1} x - \sin^{-1} x}{\sqrt{1-x^2}}$$

c.

stationary point occurs when

$$h'(x) = 0$$

$$0 = \frac{\cos^{-1} x - \sin^{-1} x}{\sqrt{1-x^2}}$$

this can only happen if

$$0 = \cos^{-1} x - \sin^{-1} x$$

$$\cos^{-1} x = \sin^{-1} x$$

$$x = \frac{1}{\sqrt{2}}$$

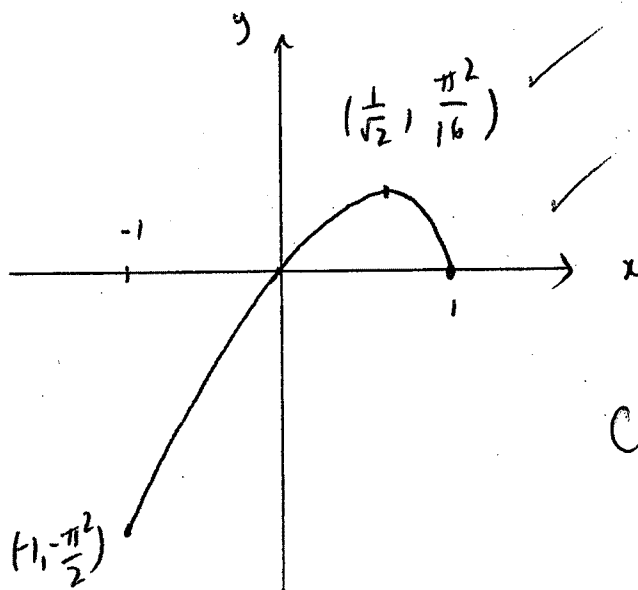
when  $\cos x = \sin x$

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{then } \frac{\pi}{4} = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \quad \frac{\pi}{4} = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$



Comments: